

Alex M Lee

Staff Data Engineer

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Skills

Programming Languages: Python, Scala, Java, SQL, NoSQL

Big Data: Airflow, Argo Workflows, Spark, Delta, Snowflake, Kafka, Debezium, dbt, Deequ, Redshift

DevOps and Cloud: Amazon Web Services (AWS), Docker, Terraform, Argo CD, Kubernetes

Experience

Airbnb

Lead Data Engineer

June 2022 – Present

- Achieved sub-10-minute latency for serving (**Snowflake**) and ingesting data (**Kafka**, **Debezium**) by building a multi-region data lake on **AWS** to provide near real-time scalable data access
- Standardized and streamlined the deployment processes and enhanced the dependability of the platform by promoting the use of **GitLab CI**, **Argo CD** and **Terraform**
- Ensured consistent data replication and integrity by designing an algorithm to simulate **MySQL** delete cascade rules starting from **CDC** events
- Produced reliable and documented data assets by introducing the CRISP-DM framework

Tecton

Data Engineer

Sep 2021 - May 2022

- Reduced issue tickets by 60% by introducing the team to test-driven development, ensuring higher quality code and the effective processing of semi-structured data with **Spark (Scala)**
- Guaranteed 100% accuracy of the company data assets by advocating for tools like **Deequ**, addressing a gap in data quality checks in **ETL** pipelines
- Increased user engagement by 30% on the mobile app by guiding data analysts and scientists in evaluating A/B test results from over 100 experiments
- Standardized machine learning deployments by leveraging BentoML to create a **CI/CD** process

Instacart

Data Engineer

Jan 2017 - May 2021

- Reduced machine failures by 15% by enabling proactive maintenance by ingesting and processing machinery data via data pipelines and creating monitoring dashboards (**Angular**, **D3.js**)
- Generated an additional **\$5** million in revenue by designing a recommendation system using **PySpark ML** to feed marketing campaigns through a reverse ETL pipeline
- Reduced support tickets by 35% by leading a team of 8 to strengthen the foundation of the data platform of a retail customer using **EMR**, **Spark**, **Delta Lake**, and **Airflow**
- Reduced bugs by 80% and lead time by 50% by mentoring the data science team of a retail customer through pair programming sessions focused on test-driven development and clean code principles
- Increased by 30% joint projects between data engineering and data science teams at a retail client, building trust and encouraging open feedback
- Reduced execution time by 50% and **AWS** expenses by 60% with legacy Spark code optimization

Education

University of Texas - Dallas - Bachelor of Computer Science, GPA: 3.8

May 2016